```
#include <stdio.h>
#include <limits.h>
typedef struct {
 int id;
 int burst;
 int deadline;
 int period;
 int remaining_burst;
 int next_deadline;
} Task;
int main() {
 int n, total_time;
 printf("1WA24CS216\n");
 printf("Enter number of tasks: ");
 scanf("%d", &n);
 Task tasks[n];
 for (int i = 0; i < n; i++) {
 printf("Enter PID, Burst, Deadline, and Period for Task %d: ", i + 1);
 scanf("%d %d %d %d", &tasks[i].id, &tasks[i].burst, &tasks[i].deadline,
&tasks[i].period);
 tasks[i].remaining_burst = 0;
 tasks[i].next_deadline = 0;
 }
 printf("Enter total scheduling time (ms): ");
 scanf("%d", &total_time);
 printf("\nScheduling occurs for %d ms\n\n", total_time);
 for (int t = 0; t < total_time; t++) {
 for (int i = 0; i < n; i++) {
 if (t % tasks[i].period == 0) {
 tasks[i].remaining_burst = tasks[i].burst;
 tasks[i].next_deadline = t + tasks[i].deadline;
```

```
}
}
int selected = -1;
int min_deadline = INT_MAX;
for (int i = 0; i < n; i++) {
if (tasks[i].remaining_burst > 0) {
if (tasks[i].next_deadline < min_deadline) {
min_deadline = tasks[i].next_deadline;
selected = i;
}
else if (tasks[i].next_deadline == min_deadline) {
if (selected == -1 || tasks[i].id < tasks[selected].id) {
selected = i;
}
}
}
}
if (selected == -1) {
printf("%dms : CPU is idle.\n", t);
} else {
printf("%dms : Task %d is running.\n", t, tasks[selected].id);
tasks[selected].remaining_burst--;
}
}
return 0;
```

}

```
1WA24CS216
Enter number of tasks: 3
Enter PID, Burst, Deadline, and Period for Task 1: 1 3 7 20
Enter PID, Burst, Deadline, and Period for Task 2: 2 2 4 5
Enter PID, Burst, Deadline, and Period for Task 3: 3 2 8 10
Enter total scheduling time (ms): 20

Scheduling occurs for 20 ms

0ms : Task 2 is running.
1ms : Task 2 is running.
2ms : Task 1 is running.
3ms : Task 1 is running.
4ms : Task 1 is running.
5ms : Task 3 is running.
6ms : Task 3 is running.
7ms : Task 2 is running.
8ms : Task 2 is running.
9ms : CPU is idle.
10ms : Task 2 is running.
11ms : Task 2 is running.
12ms : Task 3 is running.
13ms : Task 3 is running.
14ms : CPU is idle.
15ms : Task 2 is running.
16ms : Task 2 is running.
17ms : CPU is idle.
18ms : CPU is idle.
19ms : CPU is idle.

Process returned 0 (0x0)   execution time : 23.386 s
Press any key to continue.
```